

CAPE INSTITUTE OF TECHNOLOGY

(College Code: 9605)

IBM TNSDC AUG – AGENTIC AI INTERNSHIP PROJECT REPORT “AI Learning and Study Assistant”

Name : E.Suba Nandhini

Department : Computer Science and Engineering

Semester : 07

College : CAPE INSTITUTE of TECHNOLOGY

College Code :9605

Register Number :960523104095

NM ID : 9655967954802F49A97462C3C41112B5

**Programme : IBM TNSDC AUG Agentic AI Internship AI
Learning and Study Assistant**

Project Title : AI IT Helpdesk Agent

Email: Subananshini565@gmail.com

AI IT HELPDESK AGENT

Naan Mudhalvan — AI Agents Bootcamp

Final Project Report | Use Case 4

1. Abstract

This project implements an AI-powered IT Helpdesk Agent that automatically diagnoses common technical problems (Wi-Fi issues, slow systems, login failures, printer errors, crashes, and more) and provides step-by-step troubleshooting guidance. The system combines three core AI-agent capabilities — Retrieval-Augmented Generation (RAG), Tool Calling, and autonomous Agent decision-making — to behave like a real first-line IT support engineer that can also escalate unresolved or severe issues to a human team by raising a support ticket on its own.

2. Use Case Details

Field	Details
Use Case	#4 — AI IT Helpdesk Agent
What it builds	Diagnoses common technical issues and recommends troubleshooting steps using a knowledge base
Key Capabilities	Agent + RAG + Tools

3. Objective

- Build an agent that understands a user's IT problem described in plain English.
- Retrieve the most relevant solution from a structured knowledge base (RAG).
- Give the agent tools it can call on its own, like a real support system (ticket creation, status check).
- Make the agent take autonomous decisions — not just answer, but escalate when needed.

4. System Architecture

The agent follows a simple but complete Agentic pipeline:

1. User Query — the user types their IT problem in plain language.
2. RAG Retrieval — the query is converted into a word-frequency vector and compared (cosine similarity) against every article in the knowledge base to find the best match and a confidence score.
3. Agent Decision — based on the confidence score and the matched issue's severity, the agent decides one of three actions.
4. Tool Execution — if needed, the agent calls `create_support_ticket()` (simulated ITSM API) or `check_system_status()` (simulated monitoring API).
5. Response Generation — the agent returns the troubleshooting steps and/or the ticket confirmation to the user.

Decision logic used by the Agent:

- High confidence match + low/medium severity → respond with troubleshooting steps only.
- High confidence match + high severity (e.g. system crash) → respond with steps AND auto-raise a ticket for IT follow-up.
- Low confidence (no good match found) → do not guess; auto-raise a ticket for a human agent.

5. Capability Mapping

Capability	Implementation in Code	Location
RAG (Retrieval)	Cosine-similarity search over a curated IT knowledge base to retrieve the closest matching issue before responding	<code>retrieve_best_match()</code>
Tools	Callable functions simulating real systems: raising a support ticket, checking system status	<code>create_support_ticket()</code> , <code>check_system_status()</code>
Agent (Decision-making)	Chooses next action based on confidence score and issue severity: answer directly, or answer + auto-raise a ticket, or escalate when unsure	<code>HelpdeskAgent.handle_query()</code>

6. Technology Stack

- Language: Python 3 (standard library only — re, json, os, math, collections, datetime)
- Retrieval method: Bag-of-words vectorization + Cosine Similarity
- Data storage: knowledge_base.py (KB articles), tickets.json (auto-generated ticket log)
- Interface: Command-line chat loop (easily extendable to a web UI e.g. Streamlit)

7. Knowledge Base (Sample)

- Wi-Fi / Internet not connecting — severity: low
- Laptop / System running slow — severity: medium
- Forgot password / cannot login — severity: low
- Printer not printing — severity: low
- Blue screen / system crash — severity: high
- Software installation failing — severity: medium
- Email not sending/receiving — severity: low
- VPN not connecting — severity: medium

8. Sample Run / Output

You: my wifi is not connecting

Agent: [RAG match: 'Wi-Fi / Internet not connecting' | confidence=0.87 | KB-ID=KB001]

1. Restart your Wi-Fi router / modem 2. Forget & reconnect the network 3. Renew IP config 4. Check ISP if other devices also fail

You: blue screen error came suddenly

Agent: [RAG match: 'Blue screen / System crash' | confidence=0.79] ... [TOOL: create_support_ticket called] Ticket TCK1001 raised automatically.

9. Conclusion

The AI IT Helpdesk Agent successfully demonstrates the three pillars of a modern AI agent system: it retrieves relevant knowledge (RAG), interacts with external systems through callable tools, and makes autonomous decisions about when to act versus when to escalate. This mirrors how real-world enterprise IT support copilots (e.g. ServiceNow Virtual Agent, Freshservice AI) are designed, making it a strong foundation that can be extended with an LLM-based natural language generator, a larger knowledge base, and a web front-end.

10. Future Scope

- Integrate a real LLM (via API) for more natural conversational responses.
- Connect to a live ITSM tool (Freshservice / ServiceNow / Jira Service Desk) instead of a local JSON file.
- Add a Streamlit/React web interface for end users.
- Expand the knowledge base with company-specific IT policies.[github link:](#)

